

[Draft] Custom ONU module and Chassis

Custom ONU module and Chassis

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1. Overview

Meta is requesting custom XGS-PON ONU modules along with compatible chassis products to support OOB management and console requirements for Meta datacenter deployments.

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3. ONU Requirements

3.1. Common Requirements

3.1.1. All ONU Requirements

1. ONU **MUST NOT** exceed the following physical dimensions: 40mm (H) x 104mm (W) x 260mm (D).
 - a. Meta to further revise these dimensions at a later stage.
2. ONU **MUST** utilize a secure mechanical retention mechanism that supports installation in various orientations.
 - a. ONU **MUST** pass shock and vibration testing, even in downward-facing installations, without dislodging from their enclosure and without disconnecting the rear blind-mate connections.
3. ONU **MUST** have designated space on its faceplate for the Meta asset tag.
4. ONU **MUST** support XGS-PON.
5. ONU **MUST** have a user-serviceable APC fiber connector on the front.
6. ONU **MUST** support eOMCI.
7. ONU **MUST** be configured solely through eOMCI and service configuration. Active management of the ONU via other protocols/methods **MUST NOT** be required and **MUST** be disabled in production firmware images.
8. ONU **MUST** utilize input of +48VDC floating (47.4V - 51.1V), provided by a blind-mate connection on the rear of the ONU.
9. ONU power consumption **MUST NOT** exceed 25 watts.
10. ONU **MUST** operate in the following conditions: 0-40 degrees celsius; 8-85% RH non-condensing.
11. ONU **MUST** provide 16 GPIO pins on the rear of the ONU, presented to the chassis via a blind-mate connection.
 - a. The GPIO pins **MUST** utilize optoisolators for signal isolation on the pins for active components in the chassis (Pins 1-6 in the example pin out below).
 - b. GPIO pins **MAY** be on the same blind-mate connector as used for power/ground, or they **MAY** be on a separate connector.
 - c. Additional Information:
 - i. GPIO pin use cases:
 1. Health signal for active components (PSU/Fan)
 2. Chassis ID and Slot ID (4 bits/16 values)
 - a. Uses ID pins and Sense pins to provide 4 bits. **Meta to provide more detail at a later stage.**
 - ii. Example PIN out:
 1. PSU A Health Input
 2. PSU A Health Ground
 3. PSU B Health Input
 4. PSU B Health Ground

- 5. FAN Health Input
- 6. FAN Health Ground
- 7. FUTURE
- 8. FUTURE
- 9. FUTURE
- 10. Chassis ID 1
- 11. Chassis ID 2
- 12. Slot ID 1
- 13. Slot ID 2
- 14. Sense-A (Used with ID pins)
- 15. Sense-B (Used with ID pins)
- 16. Sense-C (Used with ID pins)

iii. Not all chassis variants will utilize the GPIO pins.

12. ONU **MUST** provide LED indication of Power and PON state as described below:

- a. Power
 - i. Blue solid: booted
 - ii. Green flashing: booting
 - iii. Red Green alternating: firmware updating
 - iv. Red solid: Hardware failure,
 - v. Red flashing: software failure
- b. PON
 - i. Blue solid: ranged, registered and configured (OMCI)
 - ii. Green flashing slow: ranging
 - iii. Green flashing fast: registering
 - iv. Red solid: no optical signal
 - v. Red flashing: device unauthorized

13. ONU **MUST** interoperate with OLT specified by Meta.

- a. Refer to OMCI functionality in [5.1. PON OMCI Functionality](#).
- b. Refer to OLT details in [5.2. OLT](#).

14. ONU **MUST** meet L10 compliance requirements specified by Meta.

- a. Refer to attached document: **Reference - L10-Complete-Compliance-Standard-Directives-Requirements-05042025**

15. ONU **MUST NOT** use the following components:

- a. Airoha PON SOC

16. ONU **SHOULD NOT** use the following components:

- a. Silicon Labs USB UART
- b. Molex blind mate power connector

3.1.2. Ethernet ONU Requirements

1. Ethernet ports **MUST** support the following IEEE interface standards:
 - a. 10baseT
 - b. 100baseTX
 - c. 1000baseTX
2. Ethernet ports **MUST** support autonegotiation or OMCI static configuration of interface speed and duplex.
3. Ethernet ports **MUST** support auto-MDIX.

3.1.3. Console Requirements

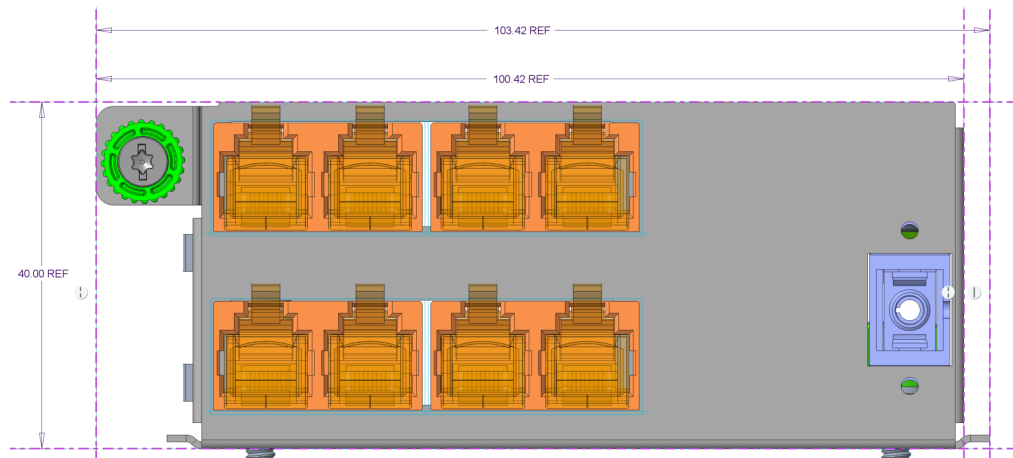
1. ONU **MUST** support exposing console sessions over SSH on user defined ports.
 - a. i.e. ports 7001, 7002, 7003 etc.
2. ONU **MUST** support multiple sessions/users utilizing the same console session simultaneously.
3. ONU **MUST** support logging to the console session for user input and system output.
4. ONU **MUST** support Secure Console requirements defined by Meta.
 - a. Refer to attached document: **Reference - Secure Console Requirements**
5. All console session communications **MUST** be supported over IPV6.
6. Console ports **MUST** support RS-232 serial interface over an 8-pin connector.
7. Console ports **MUST** support baud rates between 9600 and 115200 bps.
8. Console ports **MUST** support selectable flow control:
 - a. Hardware flow control (RTS/CTS)
 - b. Hardware flow control (DSR/DTR)
 - c. Software flow control (XON/XOFF)
 - d. No flow control
9. Console ports **MUST** support configurable stop bits.
10. Console ports **MUST** support configurable parity:
 - a. Odd
 - b. Even
 - c. Mark
 - d. Space
 - e. None
11. Console ports **MUST** support configurable data bits.

3.2. ONU Module Variants

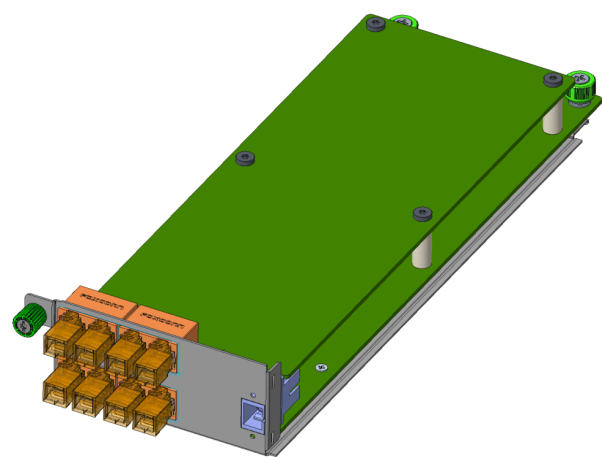
3.2.1. 4 Port Ethernet and 4 Port Console ONU Requirements

1. ONU **MUST** adhere to requirements specified in:
 - a. [3.1.1. All ONU Requirements](#)
 - b. [3.1.2. Ethernet ONU Requirements](#)
 - c. [3.1.3. Console Requirements](#)
2. ONU **MUST** provide 4 ethernet ports and 4 console ports delivered via discrete RJ45 jacks.
3. ONU **MUST** provide LED indication of carrier and speed for each ethernet port as described below:
 - a. Amber solid: Link is active at 10/100mbps speed
 - b. Green solid: Link is active at 1000mbps speed
4. ONU **MAY** provide LED indication of traffic activity for each ethernet port as described below:
 - a. Amber blinking: Data transmission at 10/100mbps speed
 - b. Green blinking: Data transmission at 1000mbps speed

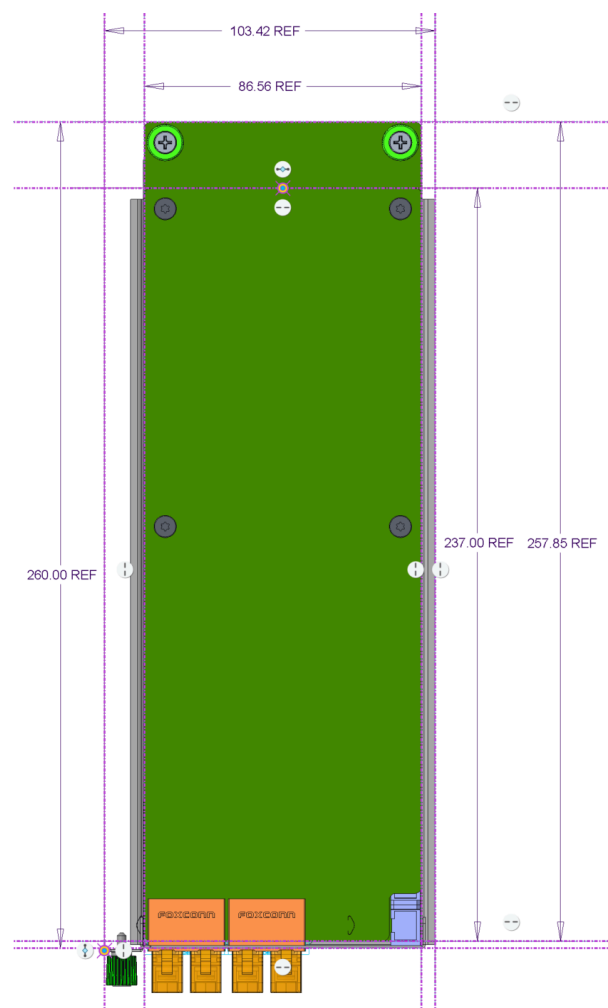
3.2.1.1. Conceptual Design - Faceplate View



3.2.1.2. Conceptual Design - ISO View



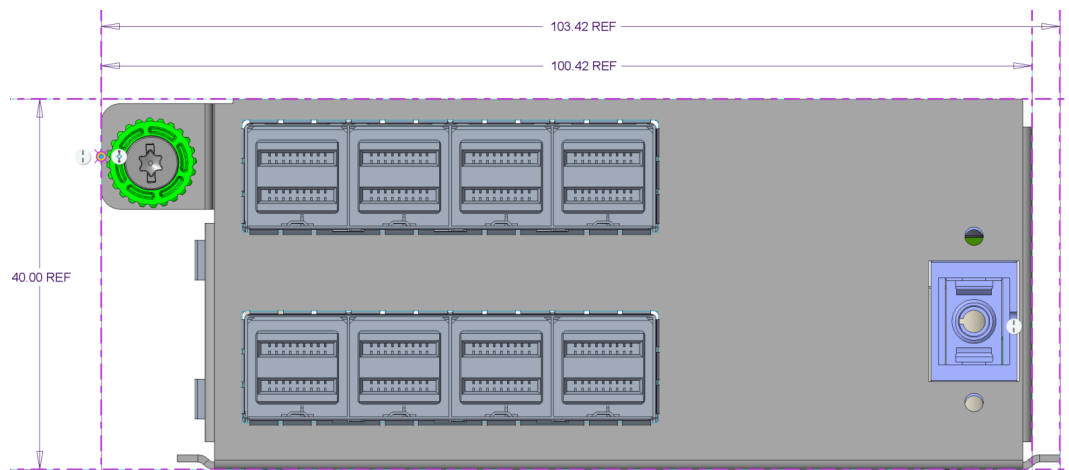
3.2.1.3. Conceptual Design - Top Down View



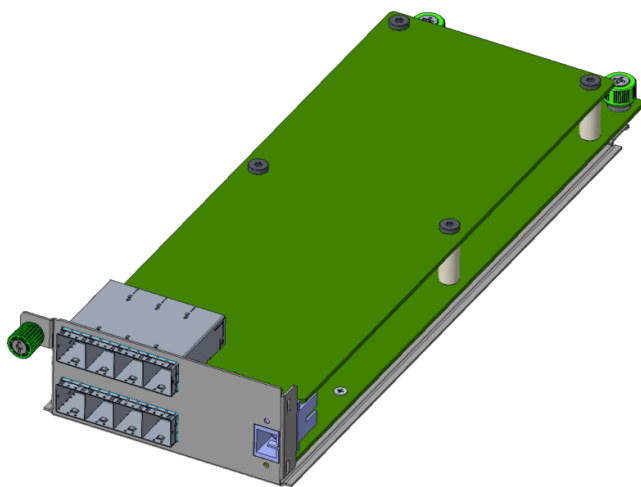
3.2.2. [Preliminary] High Density Ethernet ONU Requirements

1. ONU **MUST** adhere to requirements specified in:
 - a. [3.1.1. All ONU Requirements](#)
 - b. [3.1.2. Ethernet ONU Requirements](#)
2. ONU **MUST** provide at least 24 ethernet ports.
3. ONU **MUST** support utilizing a high density connector that breaks out to independent EIA/TIA Category 5e (or better) cables.
 - a. High density connector to be specified by Meta at a later stage.
4. ONU **SHOULD** provide LED indication of carrier and speed for each ethernet port.
5. ONU **MAY** provide LED indication of traffic activity for each ethernet port.

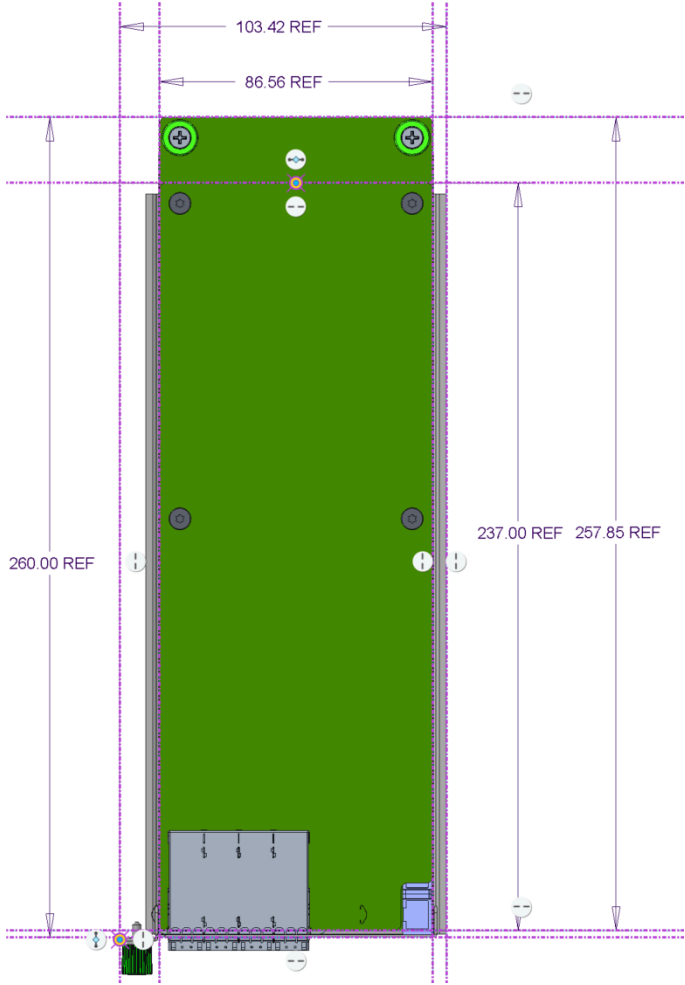
3.2.2.1. Conceptual Design - Faceplate View



3.2.2.2. Conceptual Design - ISO View



3.2.2.3. Conceptual Design - Top Down View



3.2.3. [Preliminary] High Density Console ONU Requirements

1. ONU **MUST** adhere to requirements specified in:
 - a. [3.1.1. All ONU Requirements](#)
 - b. [3.1.3. Console Requirements](#)
2. ONU **MUST** provide at least 24 console ports.
3. ONU **MUST** support utilizing a high density connector that breaks out to independent EIA/TIA Category 5e (or better) cables.
 - a. High density connector to be specified by Meta at a later stage.

4. Chassis Requirements

4.1. Common Requirements

4.1.1. All Chassis Requirements

1. Chassis **MUST** ensure adequate cooling is provided to support the simultaneous operation of all PON ONU modules.
2. Chassis **MAY** utilize ventilation holes to provide adequate cooling.
3. Chassis **MUST NOT** require active configuration nor employment of any switch, dial, or other physical control for proper operation.
4. Chassis **MUST** have designated space on its faceplate for the Meta asset tag.
5. Tools of any kind **MUST NOT** be required for the installation or removal of the PON or splitter modules.
6. Rear access to the chassis or rack **MUST NOT** be required for the installation or removal of the PON or splitter modules.
7. Chassis **MUST** have a mechanism to prevent custom ONU modules from being inserted in the wrong orientation.
8. Chassis **MUST** be tested for shock, drop, and vibration tolerance for both normal operation and shipping when empty and when fully assembled.
9. Chassis **MUST** meet L10 compliance requirements specified by Meta.
 - a. Refer to attached document: **Reference - L10-Complete-Compliance-Standard-Directives-Requirements-05042025**

4.1.2. All ORv3 Chassis Requirements

1. Chassis **MUST** extend the +48VDC powerbus from the ORv3 rack to the ONU module slots, with adequate current to support simultaneous operation ONU modules in the chassis.
2. Chassis **MUST** provide four (4) PON module slots on the front of the chassis, each compatible with the custom ONU form factor described above.
3. Chassis **MUST NOT** embed active power components within the chassis.
4. Chassis **MUST** be serviceable from the front.
5. Chassis **MUST** provide a single slot on the front of the chassis for passive splitters.
6. Chassis **MUST** support passive splitter of 40mm (H) x 51mm (W) x 150mm (D).

4.1.3. [Preliminary] All 1RU Chassis Requirements

1. Chassis **MUST** occupy 1 rack position (1RU) in the rack.
2. Chassis **MUST** include mounting ears with hole patterns appropriate for installation into a 19-inch rack using standard rack screws.
3. Chassis **MUST** support installation in various orientations and provide secure mechanical retention for all components within the chassis (i.e. ONU modules, splitter modules, power supplies)
 - a. Chassis and installed components **MUST** pass shock and vibration testing, even in downward-facing installations, without dislodging from their enclosure and without disconnecting their respective connections.
4. Chassis **SHOULD** provide hot swappable power supplies.
5. Chassis **MUST** provide a signal input to each ONU module slot (via GPIO pin) for the health status of each power supply.
6. Chassis **MUST** provide 2 (2) PON module slots on the front of the chassis, each compatible with the custom ONU form factor described above.
7. Chassis **MUST** provide a single slot on the front of the chassis for passive splitters.
 - a. Splitter specifications to be provided by Meta at a later stage.

4.2. Chassis Variants

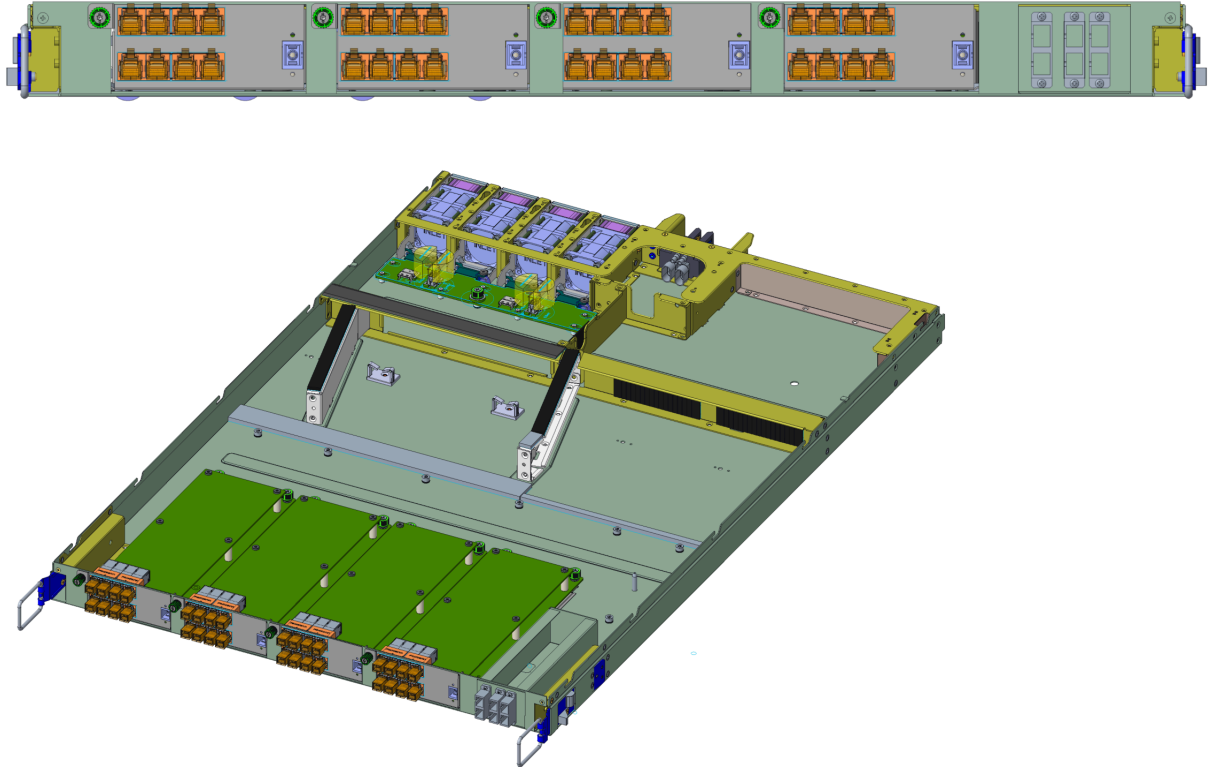
4.2.1. Above the Rack Chassis Requirements - ORv3

1. Chassis **MUST** adhere to requirements specified in:
 - a. [4.1.1. All Chassis Requirements](#)
 - b. [4.1.2. All ORv3 Chassis Requirements](#)
2. Chassis **MUST** be within the following physical dimensions: 100mm (H) x 325mm (W) x 275mm (D).
3. Chassis **MUST** be compatible with Meta specified power cable.
 - a. Refer to attached document for existing power cable specifications: **Reference - ORv3 Power Cable Spec**
 - b. Current power cable interfaces with the following component: Phoenix connector
- Part number: 1936021
 - c. This connector will interface with an existing Amphenol power cable for use in ORv3/ORv3N racks. This connector will also include grounding for the chassis and components.
4. Chassis **MUST** support tool-less attachment to the top of the ORv3 rack.
 - a. Refer to attached document: **Reference - c32629_ocp_v3**

4.2.2. [Preliminary] 1OU Chassis Requirements - ORv3

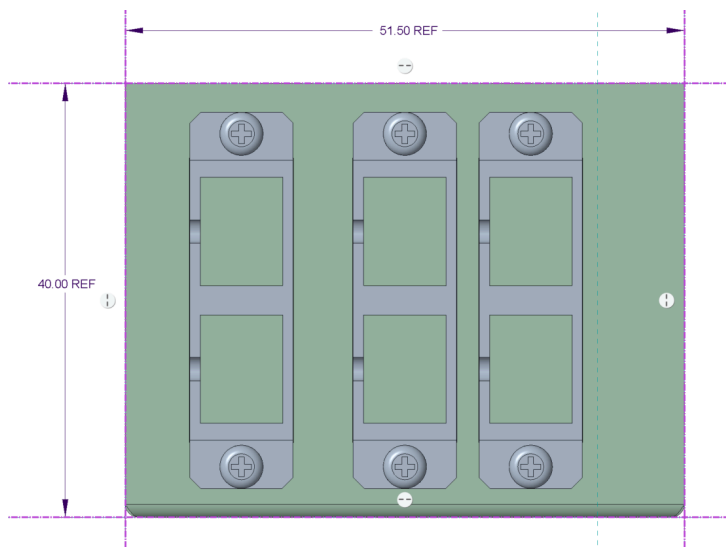
1. Chassis **MUST** adhere to requirements specified in:
 - a. [4.1.1. All Chassis Requirements](#)
 - b. [4.1.2. All ORv3 Chassis Requirements](#)
2. Chassis **MUST** fit into an ORv3 rack without additional adapters, shelves, or other hardware.
 - a. [Open Rack Base Specification Version 3](#)
 - b. [ORv3 IT Gear Design Guide](#)
 - c. [OCP Open Rack V3 IT Gear Input Connector](#)
 - d. [IT Gear Builder reference](#)
 - e. [Meta ORv3 Specification](#)
3. Chassis **MUST** occupy 1 rack position (1OU) in the rack.
 - a. Per *Open Rack V3 IT Gear Design Guide section 1.1: Design Guidelines*
 - b. Chassis height **MUST** be within 43.2mm, which is a non-standard height for ORv3 1OU. Chassis width and depth can be referenced from ORv3 specification/1OU specification.
4. Chassis **MUST** support installation and removal using the standard ORv3 shelf slide-in / slide-out mechanism for equipment defined in *Open Rack V3 IT Gear Design Guide 1.1 Design Guidelines*.
5. Chassis **MUST** directly interface to the 48VDC busbar along the rear of the rack (47.4V - 51.1V).
 - a. Per *Open Rack Base Specification Version 3 section 12.0: Electrical Requirements for IT Gear*.
6. Chassis and equipment within **MUST** be grounded via the ground bar.
 - a. Per *Open Rack Base Specification Version 3 section 12.0: Electrical Requirements for IT Gear*.
7. Chassis **MUST NOT** require any cabling connections for power or grounding.
8. Tools of any kind **MUST NOT** be required for the installation or removal of the chassis itself into or out of the rack.
9. Rear access to the chassis or rack **MUST NOT** be required for the installation or removal of the chassis itself into or out of the rack.

4.2.2.1. Conceptual Chassis Design

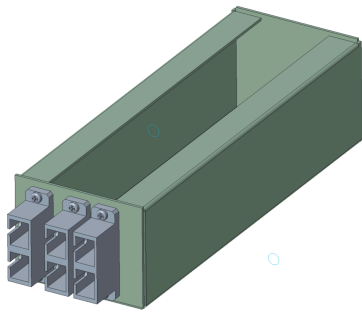


NOTE: Other components (excluding ONU modules) in the chassis are only for illustrative purposes.

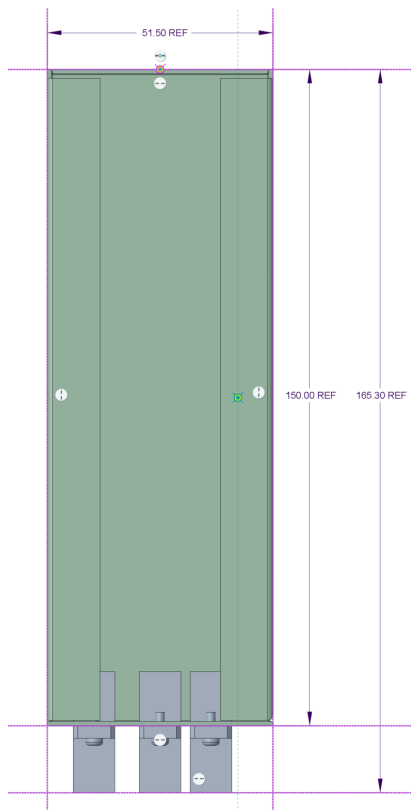
4.2.2.2. Conceptual Fiber Splitter Design - Faceplate View



4.2.2.3. Conceptual Fiber Splitter Design - ISO View



4.2.2.4. Conceptual Fiber Splitter Design - Top Down View



4.2.3. [Preliminary] 1RU Chassis - AC Variant

1. Chassis **MUST** adhere to requirements specified in:
 - a. [4.1.1. All Chassis Requirements](#)
 - b. [4.1.3. \[Preliminary\] All 1RU Chassis Requirements](#)
2. Chassis **MUST** receive dual input AC power 100-240V AC (50-60Hz) using independent and redundant power supplies.
3. Chassis **MUST** utilize IEC C13 inlets for power connections
4. Chassis **MUST** bond earth ground terminals of power connections to chassis and card safety ground planes.

4.2.4. [Preliminary] 1RU Chassis - DC Variant

1. Chassis **MUST** adhere to requirements specified in:
 - a. [4.1.1. All Chassis Requirements](#)
 - b. [4.1.3. \[Preliminary\] All 1RU Chassis Requirements](#)
2. Chassis **MUST** receive dual input DC power +/- 40-72 (floating) VDC using independent and redundant power supplies.
3. Chassis **MUST** provide a NEBS-compliant earth ground terminal near the DC power inlets.

5. Appendix

5.1. PON OMCI Functionality

5.1.1. Required ME's

NOTE: Mandatory ME's for XGS-PON are not included in this table.

ITU G.988 Clause	Name
9.1.6	Circuit pack (ANI, Ethernet UNI, RS232)
9.2.7	GAL Ethernet profile
9.2.9	FEC performance monitoring history data
9.2.13	GEM port network CTP performance monitoring history data
9.3.1	MAC bridge service profile
9.3.4	MAC bridge port configuration data (GEM, IPV6 Host)
9.3.10	IEEE 802.1p mapper service profile
9.3.11	VLAN tagging filter data
9.3.13	Extended VLAN tagging operation configuration data
9.3.30	Ethernet frame performance monitoring history data upstream
9.3.31	Ethernet frame performance monitoring history data downstream
9.4.5	IPv6 host config data
9.5.1	Physical path termination point Ethernet UNI
9.5.4	Ethernet performance monitoring history data 3
9.12.6	Threshold data 1
9.12.7	Threshold data 2
9.12.8	OMCI
9.15.1	Physical Path Termination Point RS232/RS485 UNI
9.15.2	RS232/RS485 Port Operation Configuration Data
Custom ME	SSH Server Operation
Custom ME	SSH Server Port Data

5.1.2. Example ONU Service Configuration

JSON

```
{
  "Header": {
    "Title": "5 VLANs to 5 Tconts, 4 UNI Ports, SSH Console Server to 4 RS232 ports",
    "Version": "0001",
    "Compatibility": {
    },
    "Inputs": [
      "[ONU-STATE][OLT-Service 0][UNICAST ID]",
      "[ONU-STATE][OLT-Service 1][UNICAST ID]",
      "[ONU-STATE][OLT-Service 2][UNICAST ID]",
      "[ONU-STATE][OLT-Service 3][UNICAST ID]",
      "[ONU-STATE][OLT-Service 4][UNICAST ID]",
      "[ONU-CFG][SRV-CFG Values][IPV6_ADDRESS]",
      "[ONU-CFG][SRV-CFG Values][IPV6_GATEWAY]",
      "[ONU-CFG][SRV-CFG Values][DNS_PRIMARY]",
      "[ONU-CFG][SRV-CFG Values][DNS_SECONDARY]",
      "[ONU-CFG][SRV-CFG Values][SERIAL1_ADMIN_STATE]",
      "[ONU-CFG][SRV-CFG Values][SERIAL1_BAUD_RATE]",
      "[ONU-CFG][SRV-CFG Values][SERIAL1_DATA_BITS]",
      "[ONU-CFG][SRV-CFG Values][SERIAL1_PARITY]",
      "[ONU-CFG][SRV-CFG Values][SERIAL1_STOP_BITS]",
      "[ONU-CFG][SRV-CFG Values][SERIAL1_FLOW_CONTROL]",
      "[ONU-CFG][SRV-CFG Values][SERIAL2_ADMIN_STATE]",
      "[ONU-CFG][SRV-CFG Values][SERIAL2_BAUD_RATE]",
      "[ONU-CFG][SRV-CFG Values][SERIAL2_DATA_BITS]",
      "[ONU-CFG][SRV-CFG Values][SERIAL2_PARITY]",
      "[ONU-CFG][SRV-CFG Values][SERIAL2_STOP_BITS]",
      "[ONU-CFG][SRV-CFG Values][SERIAL2_FLOW_CONTROL]",
      "[ONU-CFG][SRV-CFG Values][SERIAL3_ADMIN_STATE]",
      "[ONU-CFG][SRV-CFG Values][SERIAL3_BAUD_RATE]",
      "[ONU-CFG][SRV-CFG Values][SERIAL3_DATA_BITS]",
      "[ONU-CFG][SRV-CFG Values][SERIAL3_PARITY]",
      "[ONU-CFG][SRV-CFG Values][SERIAL3_STOP_BITS]",
      "[ONU-CFG][SRV-CFG Values][SERIAL3_FLOW_CONTROL]",
      "[ONU-CFG][SRV-CFG Values][SERIAL4_ADMIN_STATE]",
      "[ONU-CFG][SRV-CFG Values][SERIAL4_BAUD_RATE]",
      "[ONU-CFG][SRV-CFG Values][SERIAL4_DATA_BITS]",
      "[ONU-CFG][SRV-CFG Values][SERIAL4_PARITY]",
      "[ONU-CFG][SRV-CFG Values][SERIAL4_STOP_BITS]",
      "[ONU-CFG][SRV-CFG Values][SERIAL4_FLOW_CONTROL]"
    ],
  },
}
```

```
"Inputs Ext": {
  "[ONU-CFG][SRV-CFG Values][IPV6_ADDRESS]": {
    "type": "string"
  },
  "[ONU-CFG][SRV-CFG Values][IPV6_GATEWAY]": {
    "type": "string"
  },
  "[ONU-CFG][SRV-CFG Values][DNS_PRIMARY]": {
    "type": "string"
  },
  "[ONU-CFG][SRV-CFG Values][DNS_SECONDARY]": {
    "type": "string"
  },
  "[ONU-CFG][SRV-CFG Values][SERIAL1_ADMIN_STATE]": {
    "type": "string"
  },
  "[ONU-CFG][SRV-CFG Values][SERIAL1_BAUD_RATE]": {
    "type": "string"
  },
  "[ONU-CFG][SRV-CFG Values][SERIAL1_DATA_BITS]": {
    "type": "string"
  },
  "[ONU-CFG][SRV-CFG Values][SERIAL1_PARITY]": {
    "type": "string"
  },
  "[ONU-CFG][SRV-CFG Values][SERIAL1_STOP_BITS]": {
    "type": "string"
  },
  "[ONU-CFG][SRV-CFG Values][SERIAL1_FLOW_CONTROL]": {
    "type": "string"
  },
  "[ONU-CFG][SRV-CFG Values][SERIAL2_ADMIN_STATE]": {
    "type": "string"
  },
  "[ONU-CFG][SRV-CFG Values][SERIAL2_BAUD_RATE]": {
    "type": "string"
  },
  "[ONU-CFG][SRV-CFG Values][SERIAL2_DATA_BITS]": {
    "type": "string"
  },
  "[ONU-CFG][SRV-CFG Values][SERIAL2_PARITY]": {
    "type": "string"
  },
  "[ONU-CFG][SRV-CFG Values][SERIAL2_STOP_BITS]": {
```

```

        "type": "string"
    },
    "[ONU-CFG][SRV-CFG Values][SERIAL2_FLOW_CONTROL]": {
        "type": "string"
    },
    "[ONU-CFG][SRV-CFG Values][SERIAL3_ADMIN_STATE]": {
        "type": "string"
    },
    "[ONU-CFG][SRV-CFG Values][SERIAL3_BAUD_RATE]": {
        "type": "string"
    },
    "[ONU-CFG][SRV-CFG Values][SERIAL3_DATA_BITS]": {
        "type": "string"
    },
    "[ONU-CFG][SRV-CFG Values][SERIAL3_PARITY]": {
        "type": "string"
    },
    "[ONU-CFG][SRV-CFG Values][SERIAL3_STOP_BITS]": {
        "type": "string"
    },
    "[ONU-CFG][SRV-CFG Values][SERIAL3_FLOW_CONTROL]": {
        "type": "string"
    },
    "[ONU-CFG][SRV-CFG Values][SERIAL4_ADMIN_STATE]": {
        "type": "string"
    },
    "[ONU-CFG][SRV-CFG Values][SERIAL4_BAUD_RATE]": {
        "type": "string"
    },
    "[ONU-CFG][SRV-CFG Values][SERIAL4_DATA_BITS]": {
        "type": "string"
    },
    "[ONU-CFG][SRV-CFG Values][SERIAL4_PARITY]": {
        "type": "string"
    },
    "[ONU-CFG][SRV-CFG Values][SERIAL4_STOP_BITS]": {
        "type": "string"
    },
    "[ONU-CFG][SRV-CFG Values][SERIAL4_FLOW_CONTROL]": {
        "type": "string"
    }
},
"Ports": [
    "UNI-ETH 1",

```

```

        "UNI-ETH 2",
        "UNI-ETH 3",
        "UNI-ETH 4"
    ]
},
"OMCI": [
    {
        "AniG": {
            "32769": {
                "lower_optical_threshold": 255,
                "upper_optical_threshold": 255
            }
        }
    },
    {
        "CircuitPack": {
            "32": {
                "administrative_state": 0
            }
        }
    },
    {
        "GalEthernetProfile": {
            "1": {
                "max_gem_payload_size": 48
            }
        }
    },
    {
        "Tcont": {
            "32768": {
                "alloc_id": "[ONU-STATE][OLT-Service 0][UNICAST ID]"
            },
            "32769": {
                "alloc_id": "[ONU-STATE][OLT-Service 1][UNICAST ID]"
            },
            "32770": {
                "alloc_id": "[ONU-STATE][OLT-Service 2][UNICAST ID]"
            },
            "32771": {
                "alloc_id": "[ONU-STATE][OLT-Service 3][UNICAST ID]"
            },
            "32772": {
                "alloc_id": "[ONU-STATE][OLT-Service 4][UNICAST ID]"
            }
        }
    }
]

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    }
  },
  {
    "GemPortNetworkCtp": {
      "256": {
        "port_id": "[ONU-STATE][OLT-Service 0][UNICAST ID]",
        "direction": 3,
        "priority_queue_pointer_downstream": 0,
        "tcont_pointer": 32768,
        "traffic_management_pointer_upstream": 32768,
        "encryption_key_ring": 3
      },
      "257": {
        "port_id": "[ONU-STATE][OLT-Service 1][UNICAST ID]",
        "direction": 3,
        "priority_queue_pointer_downstream": 0,
        "tcont_pointer": 32769,
        "traffic_management_pointer_upstream": 32776,
        "encryption_key_ring": 3
      },
      "258": {
        "port_id": "[ONU-STATE][OLT-Service 2][UNICAST ID]",
        "direction": 3,
        "priority_queue_pointer_downstream": 0,
        "tcont_pointer": 32770,
        "traffic_management_pointer_upstream": 32784,
        "encryption_key_ring": 3
      },
      "259": {
        "port_id": "[ONU-STATE][OLT-Service 3][UNICAST ID]",
        "direction": 3,
        "priority_queue_pointer_downstream": 0,
        "tcont_pointer": 32771,
        "traffic_management_pointer_upstream": 32792,
        "encryption_key_ring": 3
      },
      "260": {
        "port_id": "[ONU-STATE][OLT-Service 4][UNICAST ID]",
        "direction": 3,
        "priority_queue_pointer_downstream": 0,
        "tcont_pointer": 32772,
        "traffic_management_pointer_upstream": 32800,
        "encryption_key_ring": 3
      }
    }
  }
}

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    }
  },
  {
    "MacBridgeServiceProfile": {
      "513": {
        "dynamic_filtering_ageing_time": 0,
        "forward_delay": 3840,
        "hello_time": 512,
        "learning_ind": 0,
        "mac_learning_depth": 0,
        "max_age": 5120,
        "port_bridging_ind": 0,
        "priority": 32768,
        "spanning_tree_ind": 0,
        "unknown_mac_address_discard": 0
      },
      "514": {
        "dynamic_filtering_ageing_time": 0,
        "forward_delay": 3840,
        "hello_time": 512,
        "learning_ind": 0,
        "mac_learning_depth": 0,
        "max_age": 5120,
        "port_bridging_ind": 0,
        "priority": 32768,
        "spanning_tree_ind": 0,
        "unknown_mac_address_discard": 0
      },
      "515": {
        "dynamic_filtering_ageing_time": 0,
        "forward_delay": 3840,
        "hello_time": 512,
        "learning_ind": 0,
        "mac_learning_depth": 0,
        "max_age": 5120,
        "port_bridging_ind": 0,
        "priority": 32768,
        "spanning_tree_ind": 0,
        "unknown_mac_address_discard": 0
      },
      "516": {
        "dynamic_filtering_ageing_time": 0,
        "forward_delay": 3840,

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        "hello_time": 512,
        "learning_ind": 0,
        "mac_learning_depth": 0,
        "max_age": 5120,
        "port_bridging_ind": 0,
        "priority": 32768,
        "spanning_tree_ind": 0,
        "unknown_mac_address_discard": 0
    },
    "517": {
        "dynamic_filtering_ageing_time": 0,
        "forward_delay": 3840,
        "hello_time": 512,
        "learning_ind": 0,
        "mac_learning_depth": 0,
        "max_age": 5120,
        "port_bridging_ind": 0,
        "priority": 32768,
        "spanning_tree_ind": 0,
        "unknown_mac_address_discard": 0
    }
},
{
    "GemInterworkingTp": {
        "32768": {
            "gem_port_network_ctp_pointer": 256,
            "interworking_option": 5,
            "service_profile_pointer": 32768,
            "interworking_tp_pointer": 0,
            "gal_profile_pointer": 1
        },
        "32769": {
            "gem_port_network_ctp_pointer": 257,
            "interworking_option": 5,
            "service_profile_pointer": 32769,
            "interworking_tp_pointer": 0,
            "gal_profile_pointer": 1
        },
        "32770": {
            "gem_port_network_ctp_pointer": 258,
            "interworking_option": 5,
            "service_profile_pointer": 32770,
            "interworking_tp_pointer": 0,

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        "gal_profile_pointer": 1
    },
    "32771": {
        "gem_port_network_ctp_pointer": 259,
        "interworking_option": 5,
        "service_profile_pointer": 32771,
        "interworking_tp_pointer": 0,
        "gal_profile_pointer": 1
    },
    "32772": {
        "gem_port_network_ctp_pointer": 260,
        "interworking_option": 5,
        "service_profile_pointer": 32772,
        "interworking_tp_pointer": 0,
        "gal_profile_pointer": 1
    }
}
},
{
    "Ieee8021pMapperServiceProfile": {
        "32768": {
            "tp_pointer": 65535,
            "unmarked_frame_option": 0,
            "default_p_bit_assumption": 0,
            "tp_type": 0,
            "interwork_tp_pointer_for_p_bit_priority_0": 32768,
            "interwork_tp_pointer_for_p_bit_priority_1": 32768,
            "interwork_tp_pointer_for_p_bit_priority_2": 32768,
            "interwork_tp_pointer_for_p_bit_priority_3": 32768,
            "interwork_tp_pointer_for_p_bit_priority_4": 32768,
            "interwork_tp_pointer_for_p_bit_priority_5": 32768,
            "interwork_tp_pointer_for_p_bit_priority_6": 32768,
            "interwork_tp_pointer_for_p_bit_priority_7": 32768
        },
        "32769": {
            "tp_pointer": 65535,
            "unmarked_frame_option": 0,
            "default_p_bit_assumption": 0,
            "tp_type": 0,
            "interwork_tp_pointer_for_p_bit_priority_0": 32769,
            "interwork_tp_pointer_for_p_bit_priority_1": 32769,
            "interwork_tp_pointer_for_p_bit_priority_2": 32769,
            "interwork_tp_pointer_for_p_bit_priority_3": 32769,
            "interwork_tp_pointer_for_p_bit_priority_4": 32769,

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        "interwork_tp_pointer_for_p_bit_priority_5": 32769,
        "interwork_tp_pointer_for_p_bit_priority_6": 32769,
        "interwork_tp_pointer_for_p_bit_priority_7": 32769
    },
    "32770": {
        "tp_pointer": 65535,
        "unmarked_frame_option": 0,
        "default_p_bit_assumption": 0,
        "tp_type": 0,
        "interwork_tp_pointer_for_p_bit_priority_0": 32770,
        "interwork_tp_pointer_for_p_bit_priority_1": 32770,
        "interwork_tp_pointer_for_p_bit_priority_2": 32770,
        "interwork_tp_pointer_for_p_bit_priority_3": 32770,
        "interwork_tp_pointer_for_p_bit_priority_4": 32770,
        "interwork_tp_pointer_for_p_bit_priority_5": 32770,
        "interwork_tp_pointer_for_p_bit_priority_6": 32770,
        "interwork_tp_pointer_for_p_bit_priority_7": 32770
    },
    "32771": {
        "tp_pointer": 65535,
        "unmarked_frame_option": 0,
        "default_p_bit_assumption": 0,
        "tp_type": 0,
        "interwork_tp_pointer_for_p_bit_priority_0": 32771,
        "interwork_tp_pointer_for_p_bit_priority_1": 32771,
        "interwork_tp_pointer_for_p_bit_priority_2": 32771,
        "interwork_tp_pointer_for_p_bit_priority_3": 32771,
        "interwork_tp_pointer_for_p_bit_priority_4": 32771,
        "interwork_tp_pointer_for_p_bit_priority_5": 32771,
        "interwork_tp_pointer_for_p_bit_priority_6": 32771,
        "interwork_tp_pointer_for_p_bit_priority_7": 32771
    },
    "32772": {
        "tp_pointer": 65535,
        "unmarked_frame_option": 0,
        "default_p_bit_assumption": 0,
        "tp_type": 0,
        "interwork_tp_pointer_for_p_bit_priority_0": 32772,
        "interwork_tp_pointer_for_p_bit_priority_1": 32772,
        "interwork_tp_pointer_for_p_bit_priority_2": 32772,
        "interwork_tp_pointer_for_p_bit_priority_3": 32772,
        "interwork_tp_pointer_for_p_bit_priority_4": 32772,
        "interwork_tp_pointer_for_p_bit_priority_5": 32772,
        "interwork_tp_pointer_for_p_bit_priority_6": 32772,

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        "interwork_tp_pointer_for_p_bit_priority_7": 32772
    }
}
},
{
    "MacBridgePortConfigurationData": {
        "8449": {
            "bridge_id_pointer": 513,
            "inbound_td_pointer": 0,
            "outbound_td_pointer": 0,
            "port_num": 2,
            "port_path_cost": 1,
            "port_priority": 0,
            "tp_pointer": 32768,
            "tp_type": 3
        },
        "8450": {
            "bridge_id_pointer": 514,
            "inbound_td_pointer": 0,
            "outbound_td_pointer": 0,
            "port_num": 3,
            "port_path_cost": 1,
            "port_priority": 0,
            "tp_pointer": 32769,
            "tp_type": 3
        },
        "8451": {
            "bridge_id_pointer": 515,
            "inbound_td_pointer": 0,
            "outbound_td_pointer": 0,
            "port_num": 4,
            "port_path_cost": 1,
            "port_priority": 0,
            "tp_pointer": 32770,
            "tp_type": 3
        },
        "8452": {
            "bridge_id_pointer": 516,
            "inbound_td_pointer": 0,
            "outbound_td_pointer": 0,
            "port_num": 5,
            "port_path_cost": 1,
            "port_priority": 0,
            "tp_pointer": 32771,

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        "tp_type": 3
    },
    "8453": {
        "bridge_id_pointer": 517,
        "inbound_td_pointer": 0,
        "outbound_td_pointer": 0,
        "port_num": 6,
        "port_path_cost": 1,
        "port_priority": 0,
        "tp_pointer": 32772,
        "tp_type": 3
    }
}
},
{
    "VlanTaggingFilterData": {
        "8449": {
            "forward_operation": 16,
            "number_of_entries": 1,
            "vlan_filter_list": [
                200
            ]
        },
        "8450": {
            "forward_operation": 16,
            "number_of_entries": 1,
            "vlan_filter_list": [
                101
            ]
        },
        "8451": {
            "forward_operation": 16,
            "number_of_entries": 1,
            "vlan_filter_list": [
                102
            ]
        },
        "8452": {
            "forward_operation": 16,
            "number_of_entries": 1,
            "vlan_filter_list": [
                103
            ]
        }
    },

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        "8453": {
            "forward_operation": 16,
            "number_of_entries": 1,
            "vlan_filter_list": [
                104
            ]
        }
    },
    {
        "MacBridgePortConfigurationData": {
            "257": {
                "bridge_id_pointer": 513,
                "port_num": 12,
                "port_path_cost": 1,
                "tp_pointer": 0,
                "tp_type": 4
            },
            "258": {
                "bridge_id_pointer": 514,
                "port_num": 13,
                "port_path_cost": 1,
                "tp_pointer": 257,
                "tp_type": 1
            },
            "259": {
                "bridge_id_pointer": 515,
                "port_num": 14,
                "port_path_cost": 1,
                "tp_pointer": 258,
                "tp_type": 1
            },
            "260": {
                "bridge_id_pointer": 516,
                "port_num": 15,
                "port_path_cost": 1,
                "tp_pointer": 259,
                "tp_type": 1
            },
            "261": {
                "bridge_id_pointer": 517,
                "port_num": 16,
                "port_path_cost": 1,
                "tp_pointer": 260,
            }
        }
    }
}

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        "tp_type": 1
    }
}
},
{
    "ExtendedVlanTaggingOperationConfigurationData": {
        "257": {
            "associated_me_pointer": 0,
            "association_type": 3,
            "downstream_mode": 0,
            "input_tpid": 33024,
            "output_tpid": 33024
        },
        "258": {
            "associated_me_pointer": 257,
            "association_type": 2,
            "downstream_mode": 0,
            "input_tpid": 33024,
            "output_tpid": 33024
        },
        "259": {
            "associated_me_pointer": 258,
            "association_type": 2,
            "downstream_mode": 0,
            "input_tpid": 33024,
            "output_tpid": 33024
        },
        "260": {
            "associated_me_pointer": 259,
            "association_type": 2,
            "downstream_mode": 0,
            "input_tpid": 33024,
            "output_tpid": 33024
        },
        "261": {
            "associated_me_pointer": 260,
            "association_type": 2,
            "downstream_mode": 0,
            "input_tpid": 33024,
            "output_tpid": 33024
        }
    }
}
},
{

```

```
"ExtendedVlanTaggingOperationConfigurationData": {
  "257": {
    "received_frame_vlan_tagging_operation_table": [
      {
        "filter_ether_type": 0,
        "filter_inner_priority": 15,
        "filter_inner_tpid_de": 0,
        "filter_inner_vid": 4096,
        "filter_outer_priority": 15,
        "filter_outer_tpid_de": 0,
        "filter_outer_vid": 4096,
        "pad1": 0,
        "pad2": 0,
        "pad3": 0,
        "pad4": 0,
        "treatment_inner_priority": 0,
        "treatment_inner_tpid_de": 4,
        "treatment_inner_vid": 200,
        "treatment_outer_priority": 15,
        "treatment_outer_tpid_de": 0,
        "treatment_outer_vid": 0,
        "treatment_tags_to_remove": 0
      },
      {
        "filter_ether_type": 0,
        "filter_inner_priority": 14,
        "filter_inner_tpid_de": 0,
        "filter_inner_vid": 4096,
        "filter_outer_priority": 14,
        "filter_outer_tpid_de": 0,
        "filter_outer_vid": 4096,
        "pad1": 0,
        "pad2": 0,
        "pad3": 0,
        "pad4": 0,
        "treatment_inner_priority": 15,
        "treatment_inner_tpid_de": 0,
        "treatment_inner_vid": 0,
        "treatment_outer_priority": 15,
        "treatment_outer_tpid_de": 0,
        "treatment_outer_vid": 0,
        "treatment_tags_to_remove": 0
      }
    ]
  }
}
```

```

        "filter_ether_type": 0,
        "filter_inner_priority": 14,
        "filter_inner_tpid_de": 0,
        "filter_inner_vid": 4096,
        "filter_outer_priority": 15,
        "filter_outer_tpid_de": 0,
        "filter_outer_vid": 4096,
        "pad1": 0,
        "pad2": 0,
        "pad3": 0,
        "pad4": 0,
        "treatment_inner_priority": 15,
        "treatment_inner_tpid_de": 0,
        "treatment_inner_vid": 0,
        "treatment_outer_priority": 15,
        "treatment_outer_tpid_de": 0,
        "treatment_outer_vid": 0,
        "treatment_tags_to_remove": 0
    }
]
},
"258": {
    "received_frame_vlan_tagging_operation_table": [
        {
            "filter_ether_type": 0,
            "filter_inner_priority": 15,
            "filter_inner_tpid_de": 0,
            "filter_inner_vid": 4096,
            "filter_outer_priority": 15,
            "filter_outer_tpid_de": 0,
            "filter_outer_vid": 4096,
            "pad1": 0,
            "pad2": 0,
            "pad3": 0,
            "pad4": 0,
            "treatment_inner_priority": 0,
            "treatment_inner_tpid_de": 4,
            "treatment_inner_vid": 101,
            "treatment_outer_priority": 15,
            "treatment_outer_tpid_de": 0,
            "treatment_outer_vid": 0,
            "treatment_tags_to_remove": 0
        },
        {

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        "filter_ether_type": 0,
        "filter_inner_priority": 14,
        "filter_inner_tpid_de": 0,
        "filter_inner_vid": 4096,
        "filter_outer_priority": 14,
        "filter_outer_tpid_de": 0,
        "filter_outer_vid": 4096,
        "pad1": 0,
        "pad2": 0,
        "pad3": 0,
        "pad4": 0,
        "treatment_inner_priority": 15,
        "treatment_inner_tpid_de": 0,
        "treatment_inner_vid": 0,
        "treatment_outer_priority": 15,
        "treatment_outer_tpid_de": 0,
        "treatment_outer_vid": 0,
        "treatment_tags_to_remove": 0
    },
    {
        "filter_ether_type": 0,
        "filter_inner_priority": 14,
        "filter_inner_tpid_de": 0,
        "filter_inner_vid": 4096,
        "filter_outer_priority": 15,
        "filter_outer_tpid_de": 0,
        "filter_outer_vid": 4096,
        "pad1": 0,
        "pad2": 0,
        "pad3": 0,
        "pad4": 0,
        "treatment_inner_priority": 15,
        "treatment_inner_tpid_de": 0,
        "treatment_inner_vid": 0,
        "treatment_outer_priority": 15,
        "treatment_outer_tpid_de": 0,
        "treatment_outer_vid": 0,
        "treatment_tags_to_remove": 0
    }
]
},
"259": {
    "received_frame_vlan_tagging_operation_table": [
        {

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        "filter_ether_type": 0,
        "filter_inner_priority": 15,
        "filter_inner_tpid_de": 0,
        "filter_inner_vid": 4096,
        "filter_outer_priority": 15,
        "filter_outer_tpid_de": 0,
        "filter_outer_vid": 4096,
        "pad1": 0,
        "pad2": 0,
        "pad3": 0,
        "pad4": 0,
        "treatment_inner_priority": 0,
        "treatment_inner_tpid_de": 4,
        "treatment_inner_vid": 102,
        "treatment_outer_priority": 15,
        "treatment_outer_tpid_de": 0,
        "treatment_outer_vid": 0,
        "treatment_tags_to_remove": 0
    },
    {
        "filter_ether_type": 0,
        "filter_inner_priority": 14,
        "filter_inner_tpid_de": 0,
        "filter_inner_vid": 4096,
        "filter_outer_priority": 14,
        "filter_outer_tpid_de": 0,
        "filter_outer_vid": 4096,
        "pad1": 0,
        "pad2": 0,
        "pad3": 0,
        "pad4": 0,
        "treatment_inner_priority": 15,
        "treatment_inner_tpid_de": 0,
        "treatment_inner_vid": 0,
        "treatment_outer_priority": 15,
        "treatment_outer_tpid_de": 0,
        "treatment_outer_vid": 0,
        "treatment_tags_to_remove": 0
    },
    {
        "filter_ether_type": 0,
        "filter_inner_priority": 14,
        "filter_inner_tpid_de": 0,
        "filter_inner_vid": 4096,

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        "filter_outer_priority": 15,
        "filter_outer_tpid_de": 0,
        "filter_outer_vid": 4096,
        "pad1": 0,
        "pad2": 0,
        "pad3": 0,
        "pad4": 0,
        "treatment_inner_priority": 15,
        "treatment_inner_tpid_de": 0,
        "treatment_inner_vid": 0,
        "treatment_outer_priority": 15,
        "treatment_outer_tpid_de": 0,
        "treatment_outer_vid": 0,
        "treatment_tags_to_remove": 0
    }
}
],
},
"260": {
    "received_frame_vlan_tagging_operation_table": [
        {
            "filter_ether_type": 0,
            "filter_inner_priority": 15,
            "filter_inner_tpid_de": 0,
            "filter_inner_vid": 4096,
            "filter_outer_priority": 15,
            "filter_outer_tpid_de": 0,
            "filter_outer_vid": 4096,
            "pad1": 0,
            "pad2": 0,
            "pad3": 0,
            "pad4": 0,
            "treatment_inner_priority": 0,
            "treatment_inner_tpid_de": 4,
            "treatment_inner_vid": 103,
            "treatment_outer_priority": 15,
            "treatment_outer_tpid_de": 0,
            "treatment_outer_vid": 0,
            "treatment_tags_to_remove": 0
        },
        {
            "filter_ether_type": 0,
            "filter_inner_priority": 14,
            "filter_inner_tpid_de": 0,
            "filter_inner_vid": 4096,

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        "filter_outer_priority": 14,
        "filter_outer_tpid_de": 0,
        "filter_outer_vid": 4096,
        "pad1": 0,
        "pad2": 0,
        "pad3": 0,
        "pad4": 0,
        "treatment_inner_priority": 15,
        "treatment_inner_tpid_de": 0,
        "treatment_inner_vid": 0,
        "treatment_outer_priority": 15,
        "treatment_outer_tpid_de": 0,
        "treatment_outer_vid": 0,
        "treatment_tags_to_remove": 0
    },
    {
        "filter_ether_type": 0,
        "filter_inner_priority": 14,
        "filter_inner_tpid_de": 0,
        "filter_inner_vid": 4096,
        "filter_outer_priority": 15,
        "filter_outer_tpid_de": 0,
        "filter_outer_vid": 4096,
        "pad1": 0,
        "pad2": 0,
        "pad3": 0,
        "pad4": 0,
        "treatment_inner_priority": 15,
        "treatment_inner_tpid_de": 0,
        "treatment_inner_vid": 0,
        "treatment_outer_priority": 15,
        "treatment_outer_tpid_de": 0,
        "treatment_outer_vid": 0,
        "treatment_tags_to_remove": 0
    }
]
},
"261": {
    "received_frame_vlan_tagging_operation_table": [
        {
            "filter_ether_type": 0,
            "filter_inner_priority": 15,
            "filter_inner_tpid_de": 0,
            "filter_inner_vid": 4096,

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    "filter_outer_priority": 15,
    "filter_outer_tpid_de": 0,
    "filter_outer_vid": 4096,
    "pad1": 0,
    "pad2": 0,
    "pad3": 0,
    "pad4": 0,
    "treatment_inner_priority": 0,
    "treatment_inner_tpid_de": 4,
    "treatment_inner_vid": 104,
    "treatment_outer_priority": 15,
    "treatment_outer_tpid_de": 0,
    "treatment_outer_vid": 0,
    "treatment_tags_to_remove": 0
  },
  {
    "filter_ether_type": 0,
    "filter_inner_priority": 14,
    "filter_inner_tpid_de": 0,
    "filter_inner_vid": 4096,
    "filter_outer_priority": 14,
    "filter_outer_tpid_de": 0,
    "filter_outer_vid": 4096,
    "pad1": 0,
    "pad2": 0,
    "pad3": 0,
    "pad4": 0,
    "treatment_inner_priority": 15,
    "treatment_inner_tpid_de": 0,
    "treatment_inner_vid": 0,
    "treatment_outer_priority": 15,
    "treatment_outer_tpid_de": 0,
    "treatment_outer_vid": 0,
    "treatment_tags_to_remove": 0
  },
  {
    "filter_ether_type": 0,
    "filter_inner_priority": 14,
    "filter_inner_tpid_de": 0,
    "filter_inner_vid": 4096,
    "filter_outer_priority": 15,
    "filter_outer_tpid_de": 0,
    "filter_outer_vid": 4096,
    "pad1": 0,

```

```

        "pad2": 0,
        "pad3": 0,
        "pad4": 0,
        "treatment_inner_priority": 15,
        "treatment_inner_tpid_de": 0,
        "treatment_inner_vid": 0,
        "treatment_outer_priority": 15,
        "treatment_outer_tpid_de": 0,
        "treatment_outer_vid": 0,
        "treatment_tags_to_remove": 0
    }
}
]
}
},
{
    "Cardholder": {
        "32": {
            "expected_plugin_unit_type": 192,
            "expected_port_count": 4
        }
    }
},
{
    "Ipv6HostConfigData": {
        "0": {
            "ip_options": 9,
            "ipv6_address": "[ONU-CFG][SRV-CFG Values][IPV6_ADDRESS]",
            "default_router": "[ONU-CFG][SRV-CFG
Values][IPV6_GATEWAY]",
            "primary_dns": "[ONU-CFG][SRV-CFG Values][DNS_PRIMARY]",
            "secondary_dns": "[ONU-CFG][SRV-CFG Values][DNS_SECONDARY]"
        }
    }
},
{
    "PptpRs232Rs485Uni": {
        "8193": {
            "administrative_state": "[ONU-CFG][SRV-CFG
Values][SERIAL1_ADMIN_STATE]",
            "port_mode": 1,
            "baud_rate": "[ONU-CFG][SRV-CFG
Values][SERIAL1_BAUD_RATE]",

```

```

        "data_bits": "[ONU-CFG][SRV-CFG
Values][SERIAL1_DATA_BITS]",
        "parity": "[ONU-CFG][SRV-CFG Values][SERIAL1_PARITY]",
        "stop_bits": "[ONU-CFG][SRV-CFG
Values][SERIAL1_STOP_BITS]",
        "flow_control": "[ONU-CFG][SRV-CFG
Values][SERIAL1_FLOW_CONTROL]"
    }
},
{
    "TcpUdpConfigData": {
        "7001": {
            "port_id": 7001,
            "protocol": 6,
            "tos_diffserv_field": 0,
            "ip_host_ptr": 0
        }
    }
},
{
    "Rs232Rs485PortOperCfgData": {
        "0": {
            "tcp_udp_ptr": 7001,
            "pptp_ptr": 8193
        }
    }
},
{
    "PptpRs232Rs485Uni": {
        "8194": {
            "administrative_state": "[ONU-CFG][SRV-CFG
Values][SERIAL2_ADMIN_STATE]",
            "port_mode": 1,
            "baud_rate": "[ONU-CFG][SRV-CFG
Values][SERIAL2_BAUD_RATE]",
            "data_bits": "[ONU-CFG][SRV-CFG
Values][SERIAL2_DATA_BITS]",
            "parity": "[ONU-CFG][SRV-CFG Values][SERIAL2_PARITY]",
            "stop_bits": "[ONU-CFG][SRV-CFG
Values][SERIAL2_STOP_BITS]",
            "flow_control": "[ONU-CFG][SRV-CFG
Values][SERIAL2_FLOW_CONTROL]"
        }
    }
}

```

```

    }
  },
  {
    "TcpUdpConfigData": {
      "7002": {
        "port_id": 7002,
        "protocol": 6,
        "tos_diffserv_field": 0,
        "ip_host_ptr": 0
      }
    }
  },
  {
    "Rs232Rs485PortOperCfgData": {
      "1": {
        "tcp_udp_ptr": 7002,
        "pptp_ptr": 8194
      }
    }
  },
  {
    "PptpRs232Rs485Uni": {
      "8195": {
        "administrative_state": "[ONU-CFG][SRV-CFG
Values][SERIAL3_ADMIN_STATE]",
        "port_mode": 1,
        "baud_rate": "[ONU-CFG][SRV-CFG
Values][SERIAL3_BAUD_RATE]",
        "data_bits": "[ONU-CFG][SRV-CFG
Values][SERIAL3_DATA_BITS]",
        "parity": "[ONU-CFG][SRV-CFG Values][SERIAL3_PARITY]",
        "stop_bits": "[ONU-CFG][SRV-CFG
Values][SERIAL3_STOP_BITS]",
        "flow_control": "[ONU-CFG][SRV-CFG
Values][SERIAL3_FLOW_CONTROL]"
      }
    }
  },
  {
    "TcpUdpConfigData": {
      "7003": {
        "port_id": 7003,
        "protocol": 6,
        "tos_diffserv_field": 0,

```

```

        "ip_host_ptr": 0
    }
}
},
{
    "Rs232Rs485Port0perCfgData": {
        "2": {
            "tcp_udp_ptr": 7003,
            "pptp_ptr": 8195
        }
    }
},
{
    "PptpRs232Rs485Uni": {
        "8196": {
            "administrative_state": "[ONU-CFG][SRV-CFG
Values][SERIAL4_ADMIN_STATE]",
            "port_mode": 1,
            "baud_rate": "[ONU-CFG][SRV-CFG
Values][SERIAL4_BAUD_RATE]",
            "data_bits": "[ONU-CFG][SRV-CFG
Values][SERIAL4_DATA_BITS]",
            "parity": "[ONU-CFG][SRV-CFG Values][SERIAL4_PARITY]",
            "stop_bits": "[ONU-CFG][SRV-CFG
Values][SERIAL4_STOP_BITS]",
            "flow_control": "[ONU-CFG][SRV-CFG
Values][SERIAL4_FLOW_CONTROL]"
        }
    }
},
{
    "TcpUdpConfigData": {
        "7004": {
            "port_id": 7004,
            "protocol": 6,
            "tos_diffserv_field": 0,
            "ip_host_ptr": 0
        }
    }
},
{
    "Rs232Rs485Port0perCfgData": {
        "3": {
            "tcp_udp_ptr": 7004,

```

```

        "pftp_ptr": 8196
    }
}
},
{
    "SshServerPortData": {
        "0": {
            "tcp_udp_ptr": 7001,
            "ssh_server_ptr": 0
        }
    }
},
{
    "SshServerPortData": {
        "1": {
            "tcp_udp_ptr": 7002,
            "ssh_server_ptr": 0
        }
    }
},
{
    "SshServerPortData": {
        "2": {
            "tcp_udp_ptr": 7003,
            "ssh_server_ptr": 0
        }
    }
},
{
    "SshServerPortData": {
        "3": {
            "tcp_udp_ptr": 7004,
            "ssh_server_ptr": 0
        }
    }
},
{
    "SshServerOperation": {
        "0": {
            "server_action": 1
        }
    }
},
{

```

```
"ThresholdDataOne": {
  "1": {
    "threshold_value_1": 100,
    "threshold_value_2": 100,
    "threshold_value_3": 100,
    "threshold_value_4": 100,
    "threshold_value_5": 100,
    "threshold_value_6": 100,
    "threshold_value_7": 100
  },
  "2": {
    "threshold_value_1": 100,
    "threshold_value_2": 100,
    "threshold_value_3": 100,
    "threshold_value_4": 100
  },
  "3": {
    "threshold_value_1": 100,
    "threshold_value_2": 100,
    "threshold_value_3": 100,
    "threshold_value_4": 100
  },
  "4": {
    "threshold_value_1": 100,
    "threshold_value_2": 100,
    "threshold_value_3": 100
  },
  "5": {
    "threshold_value_1": 10000,
    "threshold_value_2": 10000,
    "threshold_value_3": 100,
    "threshold_value_4": 100
  },
  "6": {
    "threshold_value_1": 100
  },
  "7": {
    "threshold_value_1": 100,
    "threshold_value_2": 100,
    "threshold_value_3": 100,
    "threshold_value_4": 100,
    "threshold_value_5": 100,
    "threshold_value_6": 100
  },
}
```

```

        "8": {
            "threshold_value_1": 100,
            "threshold_value_2": 100
        }
    },
    {
        "ThresholdDataTwo": {
            "1": {
                "threshold_value_8": 100,
                "threshold_value_9": 100,
                "threshold_value_10": 100,
                "threshold_value_11": 100,
                "threshold_value_12": 100,
                "threshold_value_13": 100,
                "threshold_value_14": 100
            }
        }
    },
    {
        "EthernetPerfMonHistData": {
            "257": {
                "threshold_data_half_id": 1
            },
            "258": {
                "threshold_data_half_id": 1
            },
            "259": {
                "threshold_data_half_id": 1
            },
            "260": {
                "threshold_data_half_id": 1
            }
        }
    },
    {
        "EthernetPerfMonHistData3": {
            "257": {
                "threshold_data_half_id": 2
            },
            "258": {
                "threshold_data_half_id": 2
            },
            "259": {

```

```

        "threshold_data_half_id": 2
    },
    "260": {
        "threshold_data_half_id": 2
    }
}
},
{
    "EthernetFramePerfMonHistDataUpstream": {
        "258": {
            "threshold_data_half_id": 3
        },
        "259": {
            "threshold_data_half_id": 3
        },
        "260": {
            "threshold_data_half_id": 3
        },
        "261": {
            "threshold_data_half_id": 3
        }
    }
},
{
    "EthernetFramePerfMonHistDataDownstream": {
        "258": {
            "threshold_data_half_id": 3
        },
        "259": {
            "threshold_data_half_id": 3
        },
        "260": {
            "threshold_data_half_id": 3
        },
        "261": {
            "threshold_data_half_id": 3
        }
    }
},
{
    "MacBridgePortPerfMonHistData": {
        "257": {
            "threshold_data_half_id": 4
        },

```

```

        "258": {
            "threshold_data_half_id": 4
        },
        "259": {
            "threshold_data_half_id": 4
        },
        "260": {
            "threshold_data_half_id": 4
        }
    }
},
{
    "FecPerfMonHistData": {
        "32769": {
            "threshold_data_half_id": 5
        }
    }
},
{
    "GemPortNetworkCtpPerfMonHistData": {
        "256": {
            "threshold_data_half_id": 6
        },
        "257": {
            "threshold_data_half_id": 6
        },
        "258": {
            "threshold_data_half_id": 6
        },
        "259": {
            "threshold_data_half_id": 6
        },
        "260": {
            "threshold_data_half_id": 6
        }
    }
},
{
    "XgPonDownstreamMgmtPerfMonHistData": {
        "32769": {
            "threshold_data_half_id": 8
        }
    }
},

```

```
{
  "XgPonUpstreamMgmtPerfMonHistData": {
    "32769": {
      "threshold_data_half_id": 8
    }
  }
}
```

5.2. OLT

5.2.1. Hardware

Brand: Ciena

Model: XCVR-SGPL04

5.2.2. Configuration Example

5.2.2.1. OLT Configuration

None

```
# OLT Configuration to support 2 ONU's
```

```
pon olts olt "OLT-AGG.abc1-PON02" nni-networks "0" tags "s1001.c200.c0"
pon olts olt "OLT-AGG.abc1-PON02" nni-networks "1" tags "s1001.c101.c0"
pon olts olt "OLT-AGG.abc1-PON02" nni-networks "2" tags "s1001.c102.c0"
pon olts olt "OLT-AGG.abc1-PON02" nni-networks "3" tags "s1001.c103.c0"
pon olts olt "OLT-AGG.abc1-PON02" nni-networks "4" tags "s1001.c104.c0"
pon olts olt "OLT-AGG.abc1-PON02" nni-networks "5" tags "s1002.c200.c0"
pon olts olt "OLT-AGG.abc1-PON02" nni-networks "6" tags "s1002.c101.c0"
pon olts olt "OLT-AGG.abc1-PON02" nni-networks "7" tags "s1002.c102.c0"
pon olts olt "OLT-AGG.abc1-PON02" nni-networks "8" tags "s1002.c103.c0"
pon olts olt "OLT-AGG.abc1-PON02" nni-networks "9" tags "s1002.c104.c0"
```

```
pon olts olt "OLT-AGG.abc1-PON02" onus "EXAMPLE-ONU-SN1" alloc-id-omcc 1
pon olts olt "OLT-AGG.abc1-PON02" onus "EXAMPLE-ONU-SN1" olt-service "0"
unicast-id 1154
pon olts olt "OLT-AGG.abc1-PON02" onus "EXAMPLE-ONU-SN1" olt-service "1"
unicast-id 1155
pon olts olt "OLT-AGG.abc1-PON02" onus "EXAMPLE-ONU-SN1" olt-service "2"
unicast-id 1156
pon olts olt "OLT-AGG.abc1-PON02" onus "EXAMPLE-ONU-SN1" olt-service "3"
unicast-id 1157
pon olts olt "OLT-AGG.abc1-PON02" onus "EXAMPLE-ONU-SN1" olt-service "4"
unicast-id 1158
```

```
pon olts olt "OLT-AGG.abc1-PON02" onus "EXAMPLE-ONU-SN2" alloc-id-omcc 2
pon olts olt "OLT-AGG.abc1-PON02" onus "EXAMPLE-ONU-SN2" olt-service "0"
unicast-id 1159
pon olts olt "OLT-AGG.abc1-PON02" onus "EXAMPLE-ONU-SN2" olt-service "1"
unicast-id 1160
pon olts olt "OLT-AGG.abc1-PON02" onus "EXAMPLE-ONU-SN2" olt-service "2"
unicast-id 1161
```

```
pon olts olt "OLT-AGG.abc1-PON02" onus "EXAMPLE-ONU-SN2" olt-service "3"  
unicast-id 1162  
pon olts olt "OLT-AGG.abc1-PON02" onus "EXAMPLE-ONU-SN2" olt-service "4"  
unicast-id 1163
```

5.2.2.2. ONU Configuration

None

```
# Config for onu 1
pon onus onu "ONU-1.abc1" device-id "EXAMPLE-ONU-SN1"
pon onus onu "ONU-1.abc1" olt-service "0" enable true name
"ONU-1.abc1-ONU-MGMT" sla-profile "SLA-PROFILE" tcont-service-ref none
pon onus onu "ONU-1.abc1" olt-service "0" nni-network "0" tags "s1001.c200.c0"
pon onus onu "ONU-1.abc1" olt-service "0" pon-network "0" tags "s0.c200.c0"
pon onus onu "ONU-1.abc1" olt-service "1" enable true name "ONU-1.abc1-PORT01"
sla-profile "SLA-PROFILE" tcont-service-ref none
pon onus onu "ONU-1.abc1" olt-service "1" nni-network "0" tags "s1001.c101.c0"
pon onus onu "ONU-1.abc1" olt-service "1" pon-network "0" tags "s0.c101.c0"
pon onus onu "ONU-1.abc1" olt-service "2" enable true name "ONU-1.abc1-PORT02"
sla-profile "SLA-PROFILE" tcont-service-ref none
pon onus onu "ONU-1.abc1" olt-service "2" nni-network "0" tags "s1001.c102.c0"
pon onus onu "ONU-1.abc1" olt-service "2" pon-network "0" tags "s0.c102.c0"
pon onus onu "ONU-1.abc1" olt-service "3" enable true name "ONU-1.abc1-PORT03"
sla-profile "SLA-PROFILE" tcont-service-ref none
pon onus onu "ONU-1.abc1" olt-service "3" nni-network "0" tags "s1001.c103.c0"
pon onus onu "ONU-1.abc1" olt-service "3" pon-network "0" tags "s0.c103.c0"
pon onus onu "ONU-1.abc1" olt-service "4" enable true name "ONU-1.abc1-PORT04"
sla-profile "SLA-PROFILE" tcont-service-ref none
pon onus onu "ONU-1.abc1" olt-service "4" nni-network "0" tags "s1001.c104.c0"
pon onus onu "ONU-1.abc1" olt-service "4" pon-network "0" tags "s0.c104.c0"
pon onus onu "ONU-1.abc1" olt mac-address "0" mac-address "ANY"
pon onus onu "ONU-1.abc1" onu cvlan-id 0 fw-bank-ptr 0 name "ONU-1"
pon-max-frame-size 2000 service-config "SERVICE-CONFIG.json"
pon onus onu "ONU-1.abc1" onu fw-bank-files "0" file "EXAMPLE-ONU-FW.img"
pon onus onu "ONU-1.abc1" onu fw-bank-files "1" file "EXAMPLE-ONU-FW.img"
pon onus onu "ONU-1.abc1" onu fw-bank-versions "0" version "R5.0.0"
pon onus onu "ONU-1.abc1" onu fw-bank-versions "1" version "R5.0.0"
pon onus onu "ONU-1.abc1" service-config-values "IPV6_ADDRESS" value "2001:3"
value-type str
ing
pon onus onu "ONU-1.abc1" service-config-values "IPV6_GATEWAY" value "2001::1"
value-type string
pon onus onu "ONU-1.abc1" service-config-values "DNS_PRIMARY" value "::"
value-type string
pon onus onu "ONU-1.abc1" service-config-values "DNS_SECONDARY" value "::"
value-type string
pon onus onu "ONU-1.abc1" service-config-values "SERIAL1_ADMIN_STATE" value 0
pon onus onu "ONU-1.abc1" service-config-values "SERIAL1_BAUD_RATE" value 5
pon onus onu "ONU-1.abc1" service-config-values "SERIAL1_DATA_BITS" value 8
pon onus onu "ONU-1.abc1" service-config-values "SERIAL1_PARITY" value 0
```

```
pon onus onu "ONU-1.abc1" service-config-values "SERIAL1_STOP_BITS" value 1
pon onus onu "ONU-1.abc1" service-config-values "SERIAL1_FLOW_CONTROL" value 0
pon onus onu "ONU-1.abc1" service-config-values "SERIAL2_ADMIN_STATE" value 0
pon onus onu "ONU-1.abc1" service-config-values "SERIAL2_BAUD_RATE" value 5
pon onus onu "ONU-1.abc1" service-config-values "SERIAL2_DATA_BITS" value 8
pon onus onu "ONU-1.abc1" service-config-values "SERIAL2_PARITY" value 0
pon onus onu "ONU-1.abc1" service-config-values "SERIAL2_STOP_BITS" value 1
pon onus onu "ONU-1.abc1" service-config-values "SERIAL2_FLOW_CONTROL" value 0
pon onus onu "ONU-1.abc1" service-config-values "SERIAL3_ADMIN_STATE" value 0
pon onus onu "ONU-1.abc1" service-config-values "SERIAL3_BAUD_RATE" value 5
pon onus onu "ONU-1.abc1" service-config-values "SERIAL3_DATA_BITS" value 8
pon onus onu "ONU-1.abc1" service-config-values "SERIAL3_PARITY" value 0
pon onus onu "ONU-1.abc1" service-config-values "SERIAL3_STOP_BITS" value 1
pon onus onu "ONU-1.abc1" service-config-values "SERIAL3_FLOW_CONTROL" value 0
pon onus onu "ONU-1.abc1" service-config-values "SERIAL4_ADMIN_STATE" value 0
pon onus onu "ONU-1.abc1" service-config-values "SERIAL4_BAUD_RATE" value 5
pon onus onu "ONU-1.abc1" service-config-values "SERIAL4_DATA_BITS" value 8
pon onus onu "ONU-1.abc1" service-config-values "SERIAL4_PARITY" value 0
pon onus onu "ONU-1.abc1" service-config-values "SERIAL4_STOP_BITS" value 1
pon onus onu "ONU-1.abc1" service-config-values "SERIAL4_FLOW_CONTROL" value 0
pon onus onu "ONU-1.abc1" uni "UNI-ETH 1" enable true max-frame-size 1518
pon onus onu "ONU-1.abc1" uni "UNI-ETH 2" enable true max-frame-size 1518
pon onus onu "ONU-1.abc1" uni "UNI-ETH 3" enable true max-frame-size 1518
pon onus onu "ONU-1.abc1" uni "UNI-ETH 4" enable true max-frame-size 1518
pon onus onu "ONU-2.abc1" device-id "EXAMPLE-ONU-SN2"
pon onus onu "ONU-2.abc1" olt-service "0" enable true name
"ONU-2.abc1-ONU-MGMT" sla-profile "SLA-PROFILE" tcont-service-ref none
pon onus onu "ONU-2.abc1" olt-service "0" nni-network "0" tags "s1002.c200.c0"
pon onus onu "ONU-2.abc1" olt-service "0" pon-network "0" tags "s0.c200.c0"
pon onus onu "ONU-2.abc1" olt-service "1" enable true name "ONU-2.abc1-PORT01"
sla-profile "SLA-PROFILE" tcont-service-ref none
pon onus onu "ONU-2.abc1" olt-service "1" nni-network "0" tags "s1002.c101.c0"
pon onus onu "ONU-2.abc1" olt-service "1" pon-network "0" tags "s0.c101.c0"
pon onus onu "ONU-2.abc1" olt-service "2" enable true name "ONU-2.abc1-PORT02"
sla-profile "SLA-PROFILE" tcont-service-ref none
pon onus onu "ONU-2.abc1" olt-service "2" nni-network "0" tags "s1002.c102.c0"
pon onus onu "ONU-2.abc1" olt-service "2" pon-network "0" tags "s0.c102.c0"
pon onus onu "ONU-2.abc1" olt-service "3" enable true name "ONU-2.abc1-PORT03"
sla-profile "SLA-PROFILE" tcont-service-ref none
pon onus onu "ONU-2.abc1" olt-service "3" nni-network "0" tags "s1002.c103.c0"
pon onus onu "ONU-2.abc1" olt-service "3" pon-network "0" tags "s0.c103.c0"
pon onus onu "ONU-2.abc1" olt-service "4" enable true name "ONU-2.abc1-PORT04"
sla-profile "SLA-PROFILE" tcont-service-ref none
pon onus onu "ONU-2.abc1" olt-service "4" nni-network "0" tags "s1002.c104.c0"
```

```

pon onus onu "ONU-2.abc1" olt-service "4" pon-network "0" tags "s0.c104.c0"
pon onus onu "ONU-2.abc1" olt mac-address "0" mac-address "ANY"
pon onus onu "ONU-2.abc1" onu cvlan-id 0 fw-bank-ptr 0 name "ONU-2"
pon-max-frame-size 2000 service-config "SERVICE-CONFIG.json"
pon onus onu "ONU-2.abc1" onu fw-bank-files "0" file "EXAMPLE-ONU-FW.img"
pon onus onu "ONU-2.abc1" onu fw-bank-files "1" file "EXAMPLE-ONU-FW.img"

# Config for onu 2
pon onus onu "ONU-2.abc1" onu fw-bank-versions "0" version "R5.0.0"
pon onus onu "ONU-2.abc1" onu fw-bank-versions "1" version "R5.0.0"
pon onus onu "ONU-2.abc1" service-config-values "IPV6_ADDRESS" value "2001::2"
value-type string
pon onus onu "ONU-2.abc1" service-config-values "IPV6_GATEWAY" value "2001::1"
value-type string
pon onus onu "ONU-2.abc1" service-config-values "DNS_PRIMARY" value "::"
value-type string
pon onus onu "ONU-2.abc1" service-config-values "DNS_SECONDARY" value "::"
value-type string
pon onus onu "ONU-2.abc1" service-config-values "SERIAL1_ADMIN_STATE" value 0
pon onus onu "ONU-2.abc1" service-config-values "SERIAL1_BAUD_RATE" value 5
pon onus onu "ONU-2.abc1" service-config-values "SERIAL1_DATA_BITS" value 8
pon onus onu "ONU-2.abc1" service-config-values "SERIAL1_PARITY" value 0
pon onus onu "ONU-2.abc1" service-config-values "SERIAL1_STOP_BITS" value 1
pon onus onu "ONU-2.abc1" service-config-values "SERIAL1_FLOW_CONTROL" value 0
pon onus onu "ONU-2.abc1" service-config-values "SERIAL2_ADMIN_STATE" value 0
pon onus onu "ONU-2.abc1" service-config-values "SERIAL2_BAUD_RATE" value 5
pon onus onu "ONU-2.abc1" service-config-values "SERIAL2_DATA_BITS" value 8
pon onus onu "ONU-2.abc1" service-config-values "SERIAL2_PARITY" value 0
pon onus onu "ONU-2.abc1" service-config-values "SERIAL2_STOP_BITS" value 1
pon onus onu "ONU-2.abc1" service-config-values "SERIAL2_FLOW_CONTROL" value 0
pon onus onu "ONU-2.abc1" service-config-values "SERIAL3_ADMIN_STATE" value 0
pon onus onu "ONU-2.abc1" service-config-values "SERIAL3_BAUD_RATE" value 5
pon onus onu "ONU-2.abc1" service-config-values "SERIAL3_DATA_BITS" value 8
pon onus onu "ONU-2.abc1" service-config-values "SERIAL3_PARITY" value 0
pon onus onu "ONU-2.abc1" service-config-values "SERIAL3_STOP_BITS" value 1
pon onus onu "ONU-2.abc1" service-config-values "SERIAL3_FLOW_CONTROL" value 0
pon onus onu "ONU-2.abc1" service-config-values "SERIAL4_ADMIN_STATE" value 0
pon onus onu "ONU-2.abc1" service-config-values "SERIAL4_BAUD_RATE" value 5
pon onus onu "ONU-2.abc1" service-config-values "SERIAL4_DATA_BITS" value 8
pon onus onu "ONU-2.abc1" service-config-values "SERIAL4_PARITY" value 0
pon onus onu "ONU-2.abc1" service-config-values "SERIAL4_STOP_BITS" value 1
pon onus onu "ONU-2.abc1" service-config-values "SERIAL4_FLOW_CONTROL" value 0
pon onus onu "ONU-2.abc1" uni "UNI-ETH 1" enable true max-frame-size 1518
pon onus onu "ONU-2.abc1" uni "UNI-ETH 2" enable true max-frame-size 1518

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pon onus onu "ONU-2.abc1" uni "UNI-ETH 3" enable true max-frame-size 1518  
pon onus onu "ONU-2.abc1" uni "UNI-ETH 4" enable true max-frame-size 1518
```